

# Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

July 24, 2026

# 1 Roadmap and rigour status

This lab-book is the *paper-track* of the project: it reproduces the af-validated (T0) results of the registry in `argument/` together with a thin spine of open conjectures that orients them. It is deliberately *not* a mirror of the whole registry — the non-rigorous results (inherited `proved-mod-audit`, `conjecture`, `numerical`, `heuristic`) are anchored once, honestly, in the status ledger of §23 and are never restated as theorems here. The north star is `op-classical`; its status is *open*, and its status-ledger anchor is 1. The definitions used below are the canonical shards `def-stochastic`, `def-almost-idempotent`, `def-near-positive-projection`, `def-signed-idempotent`, `def-negative-mass`, `def-height`, `def-visible-set`, `def-invisible-mass`, and `def-exposed`; this report references those identifiers and introduces no parallel definitions.

The target itself is the classical (stochastic) stability statement: there are universal constants  $\eta_0, C > 0$  such that every row-stochastic  $Q$  with  $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta \leq \eta_0$  admits a stochastic idempotent  $E$  with

$$\|Q - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\eta}, \quad (1)$$

with the exponent  $\frac{1}{2}$  sharp. This remains **open** in this repo.

The rigorous line is narrow by design. A result may be called rigorous here only when it is byte-matched to a local theorem under `refs/`, validated in an `af` workspace, or Lean-proved. At the time of this shard the registry records **29 af-validated results**. Those reproduced as full sections of this report are the signed-stochastic bridge 1; the halo-collapse family 2, 3, 4, 5, 6, 22, 23, 24, 25; the  $(EX)$ /factorization and fan-payment chain 7, 8, 9, 10, 11, 12; the argmin engine 13; the hiddenness witnesses and their consequences 14, 15, 16, 17, 18; the slab/capacity primitives 19, 20, 21; and the starvation obstruction 26. Every other registry result remains anchored only in the status ledger, with its lower-rung status left intact.

The working headline is itself status-sensitive. The inherited campaign distinguishes the realizable-family *linear law* from the worst-case  $\sqrt{\cdot}$ -envelope: along the realizable family the tight relation between negative mass  $\delta$  and height  $H$  is

$$\delta = \frac{1}{2} H \quad (\text{linear}),$$

while the quadratic form  $\delta \gtrsim H^2$  is only the worst-case envelope, binding because  $H \leq O(\tau) = O(\sqrt{\delta})$ . The  $\sqrt{\eta}$  exponent of (1) is nonetheless sharp, certified by the family `ex-hume` (anchored at 1), which is still `proved-mod-audit` here rather than one of the rigorous results. Crossing between the signed and stochastic pictures should cite the validated bridge 1 rather than re-normalising notation locally.

## 2 The signed-stochastic bridge

**Lemma 1** (`lem-classical-equiv`). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants. There is a universal  $C > 0$  such that:*

1. *(stochastic  $\rightarrow$  signed) if  $Q$  is row-stochastic with  $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta$ , then  $P = \theta(2Q - 1)$  is a signed affine retraction (**def-signed-idempotent**) satisfying  $\|P - Q\|_{\infty \rightarrow \infty} \leq C\eta$  and negative mass  $\delta \leq C\eta$ ;*
2. *(signed  $\rightarrow$  stochastic) conversely, row-normalising the positive parts  $p_i^+$  of a signed idempotent  $P$  yields a row-stochastic  $Q$  with  $\|P - Q\|_{\infty \rightarrow \infty} \leq 2\delta$  and  $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq 6\delta + 4\delta^2$ .*

Transcription note. The contract writes the map norm as  $\|\cdot\|$  without a subscript; per `CONVENTIONS.md` (b) it is the  $\infty \rightarrow \infty$  map norm  $\|\cdot\|_{\infty \rightarrow \infty}$ . The operator  $\theta$  (the signed retraction  $\theta(2Q - 1)$ ) is carried verbatim from the contract, which does not expand it.

**Registry contract** (`verbatim`).

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The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants:  $Q$  row-stochastic with  $\|Q^2 - Q\| \leq \eta$  gives  $P = \theta(2Q - 1)$  signed affine retraction with  $\|P - Q\| \leq C \eta$  and neg mass  $\delta \leq C \eta$ , and conversely row-normalising  $p_i^+$  gives  $Q$  with  $\|P - Q\| \leq 2 \delta$ ,  $\|Q^2 - Q\| \leq 6 \delta + 4 \delta^2$ .

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 29-node adversarial tree with root `validated` and clean taint. The human-readable export is `proofs/lem-classical-equiv/export.md`; the export also records the earlier missing-definition challenge and its repair by importing `def-negative-mass`.

**Role in the chain.** This bridge is the authorized way to move between `op-classical`'s stochastic formulation and the signed exact-idempotent registry. It is a dependency for the inherited simplex, rank-one, and approximate-simplex reductions, but those downstream reductions remain `proved-mod-audit` unless and until they clear an L0 gate.

### 3 Height collapse

**Observation 2** (obs-height-collapse). *Let  $P$  be an exact signed idempotent with  $0 < \delta(P) \leq \frac{1}{4}$ , nonempty visible set  $W(P)$ , and a hidden top vertex  $v$  of height  $H = \text{dist}_1(p_v, \text{conv } W)$  maximal among the rows. Then the invisible mass  $\sigma_v$  and the row negative mass  $\nu_v$  satisfy*

$$H(1 - \sigma_v) \leq \nu_v(2 + 4\delta).$$

**Registry contract** (verbatim).

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Height collapse: for an exact signed idempotent  $P$  with  $0 < \delta(P) \leq 1/4$ , nonempty visible set  $W(P)$ , and hidden top vertex  $v$  (height  $H = \text{dist}_1(p_v, \text{conv } W)$  maximal among rows), the invisible mass  $\sigma_v$  and the row negative mass  $\nu_v$  satisfy  $H * (1 - \sigma_v) \leq \nu_v * (2 + 4*\delta)$ .

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**Status.** proved; af: validated. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 19-node adversarial tree with root **validated** and taint 19/19 clean. The export is `proofs/obs-height-collapse/export.md`. Earlier compound "hence" clauses were excluded from the contract before validation; this section records only the validated inequality.

**Role in the chain.** This result explains why invisible mass alone does not finish the Kernel/ (EX) route: height is controlled only after pricing the complementary mass. A future dimension-free cap on the relevant invisible mass would combine with this inequality, but that cap is not proved here.

## 4 Mass split bookkeeping

**Lemma 3** (`lem-mass-split`). *Let  $P$  be an exact signed idempotent and  $v$  any row index. Writing  $a_j = P_{vj}$ ,  $a_j^+ = \max(a_j, 0)$ ,  $a_j^- = \max(-a_j, 0)$ , and  $\nu_v = \sum_j a_j^-$ , one has*

$$\sum_j a_j^+ = 1 + \nu_v.$$

**Registry contract** (`verbatim`).

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Mass split: for an exact signed idempotent  $P$  and any row index  $v$ , writing  $a_j = P_{vj}$ ,  $a_j^{+} = \max(a_j, 0)$ ,  $a_j^{-} = \max(-a_j, 0)$ , and  $\text{nu}_v = \sum_j a_j^{-}$ , one has  $\sum_j a_j^{+} = 1 + \text{nu}_v$ .

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a clean 9-node adversarial tree with root `validated` and taint 9/9 `clean`. The export is `proofs/lem-mass-split/export.md`.

**Role in the chain.** This is the mass bookkeeping dependency factored out of the first `lem-halo-collapse` elevation attempt. Its purpose is to keep later halo-collapse proof trees from re-deriving the same row-sum identity in multiple sibling branches.

## 5 Convex outsourcing

**Lemma 4** (`lem-residual-lower`). *Let  $C = \text{conv}\{y_1, \dots, y_N\} \subseteq \mathbb{R}^n$  be the convex hull of finitely many points, and suppose*

$$p = \sum_{j=1}^m c_j p_j + (1-s) q, \quad c_j \geq 0, \quad s = \sum_{j=1}^m c_j < 1,$$

*with  $p_j, q \in \mathbb{R}^n$  and  $\text{dist}_1(p_j, C) \leq \text{dist}_1(p, C)$  for every  $j$ . Then*

$$\text{dist}_1(p, C) \leq \text{dist}_1(q, C).$$

**Registry contract** (`verbatim`).

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Convex outsourcing: let  $C$  be the convex hull of finitely many points of  $\mathbb{R}^n$ , and suppose  $p = \sum_{j=1}^m c_j p_j + (1-s) q$  with points  $p_j, q$  in  $\mathbb{R}^n$ , coefficients  $c_j \geq 0$ ,  $s = \sum_{j=1}^m c_j < 1$ , and  $\text{dist}_1(p_j, C) \leq \text{dist}_1(p, C)$  for every  $j$ ; then  $\text{dist}_1(p, C) \leq \text{dist}_1(q, C)$ .

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records 32 validated live nodes, 36 total nodes including 4 archived nodes, root `validated`, and taint `36/36 clean`. The export is `proofs/lem-residual-lower/export.md`.

**Role in the chain.** This is a frame-free convex geometry dependency for `lem-halo-collapse`. It supplies the lower residual-distance comparison used when a hidden row is split into ordinary mass plus a residual term.

## 6 Residual distance bound

**Lemma 5** (`lem-residual-upper`). *Let  $C \subseteq \mathbb{R}^n$  be the convex hull of finitely many points, let  $b_1, \dots, b_M \geq 0$  and  $c_1, \dots, c_N \geq 0$  with  $m = \sum_j b_j - \sum_k c_k > 0$ , let  $p_j, r_k \in \mathbb{R}^n$  with*

$$q = \frac{\sum_j b_j p_j - \sum_k c_k r_k}{m},$$

*and let  $D_k \geq 0$  satisfy  $\|x - r_k\|_1 \leq D_k$  for all  $x \in C$  and each  $k$ . Then*

$$m \operatorname{dist}_1(q, C) \leq \sum_j b_j \operatorname{dist}_1(p_j, C) + \sum_k c_k D_k.$$

**Registry contract** (`verbatim`).

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Residual distance bound: let C be the convex hull of finitely many points of  $\mathbb{R}^n$ , let  $b_1..b_M, c_1..c_N \geq 0$  with  $m = \sum_j b_j - \sum_k c_k > 0$ , let  $p_j, r_k$  be points of  $\mathbb{R}^n$  with  $q = (\sum_j b_j p_j - \sum_k c_k r_k) / m$ , and let  $D_k \geq 0$  satisfy  $\|x - r_k\|_1 \leq D_k$  for all  $x$  in C and each  $k$ ; then  $m * \operatorname{dist}_1(q, C) \leq \sum_j b_j * \operatorname{dist}_1(p_j, C) + \sum_k c_k * D_k$ .

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records 49 validated live nodes, 52 total nodes including 3 archived nodes, root `validated`, and taint 52/52 `clean`. The export is `proofs/lem-residual-upper/export.md`. The tree size is above the 12-node brittleness target, so the tracked refactor warning remains honest rather than being hidden in prose.

**Role in the chain.** This is the companion frame-free convex geometry dependency for `lem-halo-collapse`. Together with 3 and 4, it packages the residual-recombination estimate needed by the halo-collapse bridge, now itself validated in 6.

## 7 Halo-robust height collapse

**Lemma 6** (lem-halo-collapse). *Let  $P$  be an exact signed idempotent with  $0 < \delta(P) \leq \frac{1}{4}$ , nonempty visible set  $W(P)$ , and hidden top vertex  $v$  of height  $H$ . Let  $\sigma$  be the invisible mass of  $v$ , let  $\sigma_g$  be the halo-robust invisible mass—the positive-coefficient mass  $v$  places on rows at  $\ell_1$  distance  $> \tau/4$  from  $\text{conv } W$ , where  $\tau = \sqrt{\delta}$ —and let  $\nu_v$  be the row negative mass. Then*

$$H(1 - \sigma_g) \leq (\sigma - \sigma_g) \frac{\tau}{4} + \nu_v(2 + 4\delta).$$

**Registry contract** (verbatim).

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Halo-robust height collapse: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty
visible set W(P), and hidden top vertex v of height H, let sigma be the invisible mass of v,
sigma_g the halo-robust invisible mass (the positive coefficient mass v places on rows at ell-1
distance > tau/4 from conv W, tau = sqrt(delta)), and nu_v the row negative mass; then H * (1 -
sigma_g) <= (sigma - sigma_g) * tau/4 + nu_v * (2 + 4*delta).
```

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**Status.** proved; af: validated. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 20-node adversarial tree, root **validated**, taint 20/20 clean (run 2 on the factored workspace). The export is `proofs/conj-halo-collapse/export.md`. The three af-validated dependencies 3, 4, and 5 are imported as externals, so the root rests only on validated or cited inputs.

**Elevation history.** Run 1 (pre-factoring) ballooned past the node cap (49 live nodes over the cap of 40) and was aborted as a structure signal, not a mathematical refutation: the mass-split book-keeping and the residual-distance estimates were being re-derived inline across siblings. Factoring those into the three registry sub-lemmas above cured it; run 2 closed at 20 nodes.

**Role in the chain.** This bridge refines the validated collapse bound 2 by pricing recipients inside the  $\tau/4$ -halo of the visible hull at their actual distance rather than the worst case  $H$ , so the bound stays non-vacuous even when the raw invisible mass exceeds one through self or halo mass (1). Together with a halo-robust cap  $\tilde{\sigma}_g \leq 1 - c$  for hidden top vertices it would give  $H = O(\tau)$ , the height cap of the Kernel Conjecture 1. That cap is OPEN: nothing in this section promotes the kernel chain.



## 8 Factorization bound

**Lemma 7** (lem-factorization). *Let  $P$  be an exact signed idempotent (a square real matrix with  $P^2 = P$  and all row sums equal to 1), let  $\delta(P) = \max_i \sum_j \max(-P_{ij}, 0)$ , let  $k = \text{rank } P$ , and let  $U = (u_1, \dots, u_k)$  be an actual-row basis of  $P$ —the rows  $p_{u_1}, \dots, p_{u_k}$  form a basis of the row space—with Gram volume  $\text{Vol}(U) \geq \frac{1}{2} \text{Vol}_{\max}(P)$ , where  $\text{Vol}_{\max}(P)$  is the maximum Gram volume over all actual-row bases of  $P$ . Define coordinates  $a_t(j)$  by  $p_j = \sum_t a_t(j) p_{u_t}$ , and for each pivot index  $s \in \{1, \dots, k\}$  set*

$$\begin{aligned} \beta_s(j) &= P_{u_s j}, & \lambda_s(j) &= 1 - a_s(j), & \mu_s(j) &= \sum_{t \neq s} \max(-a_t(j), 0), \\ \sigma_s(j) &= \sum_{t \neq s} \max(a_t(j), 0), & E_s(j) &= \max(\mu_s(j) - \lambda_s(j), 0), \\ \Phi_s(U) &= \sum_j \max(\beta_s(j), 0) E_s(j), & S_s^*(U) &= \sum_j \max(\beta_s(j), 0) (\sigma_s(j) + 2 \max(-\lambda_s(j), 0)). \end{aligned}$$

Then for every pivot  $s$ ,

$$S_s^*(U) \leq 2 \Phi_s(U) + 6 \delta(P).$$

**Registry contract (verbatim).**

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Factorization bound: let P be an exact signed idempotent (square real matrix with P^2 = P and all
row sums equal to 1), let delta(P) = max_i sum_j max(-P_ij, 0), let k = rank(P), and let U = (u_1,
..., u_k) be an actual-row basis of P (the rows p_{u_1}, ..., p_{u_k} of P form a basis of the row
space of P) whose Gram volume satisfies Vol(U) >= (1/2) * Vol_max(P), where Vol_max(P) is the
maximum Gram volume over all actual-row bases of P; define coordinates a_t(j) by p_j = sum_t a_t(j)
p_{u_t}, and for each pivot index s in {1, ..., k} set beta_s(j) = P_{u_s j}, lambda_s(j) = 1 -
a_s(j), mu_s(j) = sum_{t != s} max(-a_t(j), 0), sigma_s(j) = sum_{t != s} max(a_t(j), 0), E_s(j) =
max(mu_s(j) - lambda_s(j), 0), Phi_s(U) = sum_j max(beta_s(j), 0) * E_s(j), and S*_s(U) = sum_j
max(beta_s(j), 0) * (sigma_s(j) + 2 * max(-lambda_s(j), 0)); then for every pivot s, S*_s(U) <= 2 *
Phi_s(U) + 6 * delta(P).
```

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**Status.** proved; af: validated. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records an 11-live-node adversarial tree (12 total nodes, 1 archived), root validated, taint 12/12 clean, closed in run 1 after five rounds. The export is proofs/lem-factorization/e. The contract was narrowed to the single inequality before seeding; the inherited tightness claim for the constants (2,6) was deliberately not elevated and remains proved-mod-audit material in the registry shard body.

**Role in the chain.** This is the composition link of the (EX) program: any universal chart bound  $\max_s \Phi_s(U_0) \leq C_0 \delta(P)$  over the  $\theta = 1/2$  class now composes rigorously to the class-score constant  $C_{sf} = 2C_0 + 6$ . The (EX) input itself (1) remains a conjecture: certified numerical evidence supports a plateau near  $C_0 = 2$ , but nothing in this section promotes it.

## 9 Zero-sum triangle bound

**Lemma 8** (`lem-zerosum-triangle`). *Let  $w, v \in \mathbb{R}^d$  with  $v$  having zero coordinate sum,  $\sum_l v(l) = 0$ , and write  $n(x) = \sum_l \max(-x(l), 0)$ . Then*

$$n(w - v) \leq n(w) + n(v).$$

**Registry contract** (`verbatim`).

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Zero-sum triangle bound: let w and v be vectors in R^d with v having coordinate sum zero (sum_l
v(l) = 0); write n(x) = sum_l max(-x(l), 0); then n(w - v) <= n(w) + n(v).
```

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 10-node adversarial tree, root `validated`, taint 10/10 `clean`, closed in run 1 after three rounds. The export is `proofs/lem-zerosum-triangle/export.md`. The run-1 MISSING-fact challenge isolated the load-bearing hypothesis: the identity  $\sum_l \max(v(l), 0) = n(v)$  holds exactly because  $v$  has zero coordinate sum; for general  $v$  only the pointwise inequality survives.

**Role in the chain.** First factored dependency of 10: it was split out of the `lem-fan-payment` workspace after two balloon aborts forced a linear proof chain into cross-sibling re-derivations. Downstream `af` trees import it as a validated external instead of re-deriving the triangle step per node.

## 10 Weighted minimum bound

**Lemma 9** (`lem-weighted-min`). *Let  $p_1, \dots, p_m$  be positive reals with  $\sum_i p_i = 1$  and let  $n_1, \dots, n_m$  be real numbers. Then*

$$\min_{i \in \{1, \dots, m\}} n_i \leq \sum_i p_i n_i.$$

**Registry contract** (`verbatim`).

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Weighted minimum bound: let `p_1, ..., p_m` be positive reals with `sum_i p_i = 1` and let `n_1, ..., n_m` be real numbers; then `min over i in {1, ..., m} of n_i <= sum_i p_i * n_i`.

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records an 8-node adversarial tree, root `validated`, taint 8/8 `clean` (the build run's verifier calls were blanked by a network outage and the verify phase was resumed cleanly). The export is `proofs/lem-weighted-min/export.md`.

**Role in the chain.** Second factored dependency of 10: the index-selection step (some support point has value at most the weighted average). Textbook-adjacent, but registered as its own shard so `af` trees can import it as a validated external instead of re-deriving it across siblings.

## 11 Zero-sum fan payment

**Lemma 10** (lem-fan-payment). *Let  $(w_1, p_1), \dots, (w_m, p_m)$  be a finite family with vectors  $w_i \in \mathbb{R}^d$  and weights  $p_i > 0$  satisfying  $\sum_i p_i = 1$ , such that every  $w_i$  has zero coordinate sum,  $\sum_l w_i(l) = 0$ , and the weighted barycenter is zero,  $\sum_i p_i w_i = 0$ . Write  $n(w) = \sum_l \max(-w(l), 0)$ . Then*

$$\min_{i^* \in \{1, \dots, m\}} \sum_i p_i n(w_i - w_{i^*}) \leq 2 \sum_i p_i n(w_i).$$

**Registry contract** (verbatim).

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Zero-sum fan payment: let  $(w_1, p_1), \dots, (w_m, p_m)$  be a finite family with vectors  $w_i$  in  $\mathbb{R}^d$  and weights  $p_i > 0$  satisfying  $\sum_i p_i = 1$ , such that every  $w_i$  has coordinate sum zero ( $\sum_l w_i(l) = 0$ ) and the weighted barycenter is zero ( $\sum_i p_i w_i = 0$ ); write  $n(w) = \sum_l \max(-w(l), 0)$ ; then  $\min_{i^* \in \{1, \dots, m\}} \sum_i p_i n(w_i - w_{i^*}) \leq 2 * \sum_i p_i n(w_i)$ .

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**Status.** proved; af: validated. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 15-node adversarial tree, root **validated**, taint 15/15 **clean**, closed in run 3 on the factored workspace; it imports the **af**-validated dependencies 8 and 9 as externals. Runs 1–2, before the factoring, balloon-aborted at 39 and 47 nodes; the factoring cured the balloon (15 versus 47). The export is **proofs/lem-fan-payment/export.md**.

**Role in the chain.** The all-mass fan payment lemma is the discrete mechanism behind the certified plateau-2 constant on the reduced fan families of the arm-A campaign: in a reduced edge-fan chart the anchor-pivot score is exactly the weighted average  $\sum_i p_i n(w_i - w_*)$ , so this lemma bounds the fan-template  $\Phi/\delta$  by 2. The registry shard body additionally records an exact refutation certificate for the over-broad variant (arbitrary  $w_0$ , no zero-sum hypothesis, ratio 5); that certificate is wave material and is not elevated here.

## 12 Negative-part subadditivity

**Lemma 11** (`lem-negpart-subadditive`). *For all vectors  $x, y \in \mathbb{R}^d$ , writing  $n(w) = \sum_l \max(-w(l), 0)$ , one has*

$$n(x + y) \leq n(x) + n(y).$$

**Registry contract** (`verbatim`).

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Negative-part subadditivity: for all vectors x and y in R^d, writing n(w) = sum\_l max(-w(l), 0), one has n(x + y) <= n(x) + n(y).

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**Status.** `proved`; `af`: `validated`. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 16-node adversarial tree, root `validated`, taint 16/16 `clean`, closed in run 1. The export is `proofs/lem-negpart-subadditive/export.md`. The lemma was pre-factored *before* elevation, applying the balloon lesson from `lem-fan-payment` instead of re-learning it.

**Role in the chain.** Pre-factored dependency of 12: the D-restricted fan proof's barycenter step  $D \geq n(\sum_{i \in A} p_i w_i)$  uses exactly this pointwise subadditivity (with nonnegative homogeneity, which is definitional).

### 13 D-restricted zero-sum fan payment

**Lemma 12** (`lem-fan-payment-restricted`). *Let  $(w_1, p_1), \dots, (w_m, p_m)$  be a finite family with vectors  $w_i \in \mathbb{R}^d$  and weights  $p_i > 0$  satisfying  $\sum_i p_i = 1$ , such that every  $w_i$  has zero coordinate sum and the weighted barycenter is zero,  $\sum_i p_i w_i = 0$ . Write  $n(w) = \sum_l \max(-w(l), 0)$ , let  $w_*$  be a minimizer of  $v \mapsto \sum_i p_i n(w_i - v)$  over  $\{w_1, \dots, w_m\}$ , and set  $A = \{i : n(w_i - w_*) > 0\}$ . Then*

$$\sum_{i \in A} p_i n(w_i - w_*) \leq (2 + \sqrt{2}) \sum_{i \in A} p_i n(w_i).$$

**Registry contract** (`verbatim`).

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D-restricted zero-sum fan payment: let  $(w_1, p_1), \dots, (w_m, p_m)$  be a finite family with vectors  $w_i$  in  $\mathbb{R}^d$  and weights  $p_i > 0$  satisfying  $\sum_i p_i = 1$ , such that every  $w_i$  has coordinate sum zero and the weighted barycenter is zero ( $\sum_i p_i w_i = 0$ ); write  $n(w) = \sum_l \max(-w(l), 0)$ ; let  $w_*$  be a minimizer of  $v \mapsto \sum_i p_i n(w_i - v)$  over  $\{w_1, \dots, w_m\}$  and let  $A = \{i : n(w_i - w_*) > 0\}$ ; then  $\sum_{i \in A} p_i n(w_i - w_*) \leq (2 + \sqrt{2}) * \sum_{i \in A} p_i n(w_i)$ .

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**Status.** proved; `af`: validated. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 27-node adversarial tree, root `validated`, taint 27/27 `clean`, closed in run 1 after ten rounds; it imports the `af`-validated dependencies 8 and 11 as externals. The export is `proofs/lem-fan-payment-restricted/export.md`. Proof shape: with  $q$  the duplicate-pivot mass, horn (i) gives  $N \leq D/q$  and horn (ii) gives  $N \leq 2(1 - q)/(1 - 2q) D$  for  $q < 1/2$ ; the bounds cross at  $q_0 = 1 - 1/\sqrt{2}$  with common value  $2 + \sqrt{2}$ .

**Role in the chain.** The D-restricted refinement of 10: the denominator keeps only rows with positive distance to the selected pivot, the fan shadow of the degenerate-set restriction behind `conj-degenerate-payment`. The registry shard body records exact certificates (wave material, not elevated here) refuting constant 2 for this variant and attaining  $2 + \sqrt{2}$  in the limit, so the validated constant is sharp. The remaining obstruction toward `conj-degenerate-payment` is the lift from  $\theta_{-1/2}$  argmin rows to this fan model.

## 14 Pivot-removing max-stationarity

**Lemma 13** (lem-pivot-removing-move). *Let  $P$  be a rank-3 exact signed idempotent ( $P^2 = P$ , all row sums 1), let  $U = (u_0, u_1, u_2)$  be an actual-row chart whose rows  $p_{u_0}, p_{u_1}, p_{u_2}$  form a basis of the row space, let  $\text{Vol}_{\max}(P)$  be the maximum Gram volume over actual-row charts, and  $m_U = \text{Vol}(U)/\text{Vol}_{\max}(P)$ . Define old coordinates  $a_t(i)$  by  $p_i = \sum_t a_t(i) p_{u_t}$ , and set*

$$\beta_r(i) = P_{u_r i}, \quad \lambda_r(i) = 1 - a_r(i), \quad \mu_r(i) = \sum_{t \neq r} \max(-a_t(i), 0),$$

$$E_r(i) = \max(\mu_r(i) - \lambda_r(i), 0), \quad \Phi_r(U) = \sum_i \max(\beta_r(i), 0) E_r(i), \quad \Phi(U) = \max_r \Phi_r(U).$$

*Assume  $U$  is a  $\theta - \frac{1}{2}$   $\Phi$ -argmin:  $m_U \geq \frac{1}{2}$  and  $\Phi(U)$  is minimal among actual-row charts  $W$  with  $\text{Vol}(W)/\text{Vol}_{\max}(P) \geq \frac{1}{2}$ . Fix a maximal pivot  $s$  with  $\Phi_s(U) = \Phi(U)$ , and let  $j \notin \{u_0, u_1, u_2\}$  have  $c = a_s(j) \neq 0$  such that the pivot-removing chart  $V_j = U - u_s + j$  is  $\theta - \frac{1}{2}$  admissible—equivalently  $|a_s(j)| m_U \geq \frac{1}{2}$ , with volume factor  $\text{Vol}(V_j)/\text{Vol}(U) = |a_s(j)|$ . Defining new coordinates on  $V_j$  by*

$$a_s^j(i) = \frac{a_s(i)}{c}, \quad a_t^j(i) = a_t(i) - \frac{a_s(i) a_t(j)}{c} \quad (t \neq s),$$

$$E_r^j(i) = \max\left(\sum_{q \neq r} \max(-a_q^j(i), 0) - (1 - a_r^j(i)), 0\right),$$

$$\Psi_j = \Phi_s(V_j) = \sum_i \max(P_{ji}, 0) E_s^j(i), \quad \Gamma_j = \max_{r \neq s} \Phi_r(V_j) = \max_{r \neq s} \sum_i \max(P_{u_r i}, 0) E_r^j(i),$$

one has

$$\Phi_s(U) \leq \max(\Psi_j, \Gamma_j).$$

**Registry contract (verbatim).**

---

Pivot-removing max-stationarity: let  $P$  be a rank-3 exact signed idempotent (square real matrix with  $P^2 = P$  and all row sums equal to 1), let  $U = (u_0, u_1, u_2)$  be an actual-row chart whose rows  $p_{u_0}, p_{u_1}, p_{u_2}$  form a basis of the row space, let  $\text{Vol}_{\max}(P)$  be the maximum Gram volume over actual-row charts,  $m_U = \text{Vol}(U)/\text{Vol}_{\max}(P)$ , define old coordinates  $a_t(i)$  by  $p_i = \sum_t a_t(i) p_{u_t}$ ,  $\beta_r(i) = P_{u_r i}$ ,  $\lambda_r(i) = 1 - a_r(i)$ ,  $\mu_r(i) = \sum_{t \neq r} \max(-a_t(i), 0)$ ,  $E_r(i) = \max(\mu_r(i) - \lambda_r(i), 0)$ ,  $\Phi_r(U) = \sum_i \max(\beta_r(i), 0) E_r(i)$ , and  $\Phi(U) = \max_r \Phi_r(U)$ , assume  $U$  is a  $\theta$ -half  $\Phi$ -argmin meaning  $m_U \geq 1/2$  and  $\Phi(U)$  is minimal among actual-row charts  $W$  with  $\text{Vol}(W)/\text{Vol}_{\max}(P) \geq 1/2$ , fix a maximal pivot  $s$  with  $\Phi_s(U) = \Phi(U)$ , and let  $j \notin \{u_0, u_1, u_2\}$  have  $c = a_s(j) \neq 0$  such that the pivot-removing chart  $V_j = U - u_s + j$  is  $\theta$ -half admissible, equivalently  $|a_s(j)| m_U \geq 1/2$  with volume factor  $\text{Vol}(V_j)/\text{Vol}(U) = |a_s(j)|$ ; defining new coordinates on  $V_j$  by  $a_s^j(i) = a_s(i)/c$  and  $a_t^j(i) = a_t(i) - a_s(i) a_t(j)/c$  for  $t \neq s$ ,  $E_r^j(i) = \max(\sum_{q \neq r} \max(-a_q^j(i), 0) - (1 - a_r^j(i)), 0)$ ,  $\Psi_j = \Phi_s(V_j) = \sum_i \max(P_{ji}, 0) E_s^j(i)$ , and  $\Gamma_j = \max_{r \neq s} \Phi_r(V_j) = \max_{r \neq s} \sum_i \max(P_{u_r i}, 0) E_r^j(i)$ , one has  $\Phi_s(U) \leq \max(\Psi_j, \Gamma_j)$ .

---

**Status.** proved; af: validated. This is one of the rigorous in-repo results.

**Proof tree.** The registry shard records a 9-node adversarial tree, root validated, taint 9/9 clean, closed in run 1 after three rounds with zero challenges. The export is `proofs/lem-pivot-removing-move/export.r`. The statement was codified from the arm-G wave record with a fully-inline contract and elevated pre-factored, so the tree stayed small.

**Role in the chain.** This is the minimality tool of the argmin engine: it is the first statement of the campaign that genuinely uses  $\Phi$ -minimality of the chart, via the pivot-removing move with exact volume factor  $|a_s(j)|$ . Every attack on the remaining self-support control (`conj-sc` and its pivot-removing blocker reduction) consumes this disjunction: an admissible pivot-removing move must be blocked by the new-pivot score  $\Psi_j$  or by collateral  $\Gamma_j$  on another pivot. The wave-9 record realizes the volume-inadmissible and  $\Psi$ -blocked alternatives by exact certified instances; the collateral alternative is the open branch, and its analysis now rests on this validated statement.



## 15 The hiddenness dual witness and its always-tight support

Throughout this section  $P$  is an exact signed idempotent (`def-signed-idempotent`) with  $\delta = \delta(P) > 0$ , and we write the derived scales

$$\tau = \sqrt{\delta}, \quad \rho = 4\tau, \quad \kappa = \frac{1}{4}\tau,$$

as fixed in `CONVENTIONS.md` (c). For a row vertex  $v$  with row point  $p_v$  set  $d_i = p_i - p_v$  and let

$$F_v = \{j : \|p_j - p_v\|_1 \geq \rho\}$$

be the  $\rho$ -far row-index set. Recall (`def-exposed`) that the *exposedness margin* of  $v$  is

$$t^*(v) = \sup_h \min_{f \in F_v} h(p_f), \quad (2)$$

the supremum over affine *admissible exposers*  $h$  with  $h(p_v) = 0$  and  $0 \leq h(p_i) \leq 1$  for every row  $i$ ; the vertex  $v$  is *hidden* when it is not  $(\rho, \kappa)$ -exposed, i.e.  $t^*(v) < \kappa$ .

### 15.1 Existence of the witness

**Lemma 14** (`lem-hiddenness-dual-witness`). *Let  $v$  be a hidden row vertex of  $P$ . Then  $F_v \neq \emptyset$  and there exist coefficients  $\lambda_f \geq 0$  ( $f \in F_v$ ) with  $\sum_f \lambda_f = 1$ , and  $\alpha_i, \beta_i \geq 0$  over all row indices  $i$  with  $\sum_i \beta_i = t^*(v) < \kappa$ , such that*

$$\sum_{f \in F_v} \lambda_f (p_f - p_v) + \sum_i \alpha_i (p_i - p_v) = \sum_i \beta_i (p_i - p_v). \quad (3)$$

**Registry contract** (`verbatim`).

---

Hiddenness dual witness: for an exact signed idempotent  $P$  and a hidden row vertex  $v$  ( $\rho = 4 \cdot \tau$ ,  $\kappa = \tau/4$ ,  $\tau = \sqrt{\delta(P)}$ ), writing  $F_v = \{j : \|p_j - p_v\|_1 \geq \rho\}$  for the  $\rho$ -far row-index set (nonempty for hidden  $v$ ), there exist  $\lambda_f \geq 0$  ( $f \in F_v$ ) with  $\sum_f \lambda_f = 1$  and  $\alpha_i, \beta_i \geq 0$  (over all row indices  $i$ ) with  $\sum_i \beta_i = t^*(v) < \kappa$ , such that  $\sum_f \lambda_f (p_f - p_v) + \sum_i \alpha_i (p_i - p_v) = \sum_i \beta_i (p_i - p_v)$ .

---

**Proof sketch.** Hiddenness first supplies the scalar prerequisites: were  $F_v$  empty, `def-exposed` would give  $t^*(v) = +\infty$ , so a hidden vertex forces  $F_v \neq \emptyset$  and  $t^*(v) < \kappa$ . Writing an admissible exposer as  $h(x) = u \cdot (x - p_v)$ , the margin (2) is the value of the finite linear program

$$\max_{u, t} \quad t \quad \text{s.t.} \quad 0 \leq u \cdot d_i \leq 1 \quad (\forall i), \quad t \leq u \cdot d_f \quad (\forall f \in F_v), \quad (4)$$

which is feasible at  $(u, t) = (0, 0)$  and, since  $F_v \neq \emptyset$ , bounded above by 1. Dualising (4) with multipliers  $\beta_i \geq 0$  on  $u \cdot d_i \leq 1$ ,  $\alpha_i \geq 0$  on  $-u \cdot d_i \leq 0$ , and  $\lambda_f \geq 0$  on  $t - u \cdot d_f \leq 0$  gives

$$\min \sum_i \beta_i \quad \text{s.t.} \quad \sum_f \lambda_f = 1, \quad \sum_f \lambda_f d_f + \sum_i \alpha_i d_i = \sum_i \beta_i d_i.$$

Finite LP strong duality and dual attainment produce an optimal dual point with  $\sum_i \beta_i = t^*(v)$ ; substituting  $d_i = p_i - p_v$  is exactly (3), and  $t^*(v) < \kappa$  closes the count.  $\square$

**Provenance. status:** `proved`, `af: validated`; workspace `proofs/lem-hiddenness-dual-witness` (export node 1). First-principles finite LP duality with no imports; a fresh-codex hostile verifier re-derived feasibility, boundedness, attainment, and strong duality from scratch and checked the witness on the banked  $7 \times 7$  duplicate-split fixture ( $t^*(v) = \frac{1}{21} < \frac{1}{16}$ ).

## 15.2 The witness support is always tight

Fix a hidden geometrically distinct row vertex  $u$  with nonempty visible set  $W(P)$ . On the primal optimal face of (4) define the *always-tight* constraint families

$$T = \{f \in F_u : h(p_f) = t^*(u) \forall \text{ optima } h\}, \quad O = \{i : h(p_i) = 1 \forall \text{ optima}\}, \quad Z = \{i : h(p_i) = 0 \forall \text{ optima}\}.$$

**Lemma 15** (`lem-always-tight-dual-support`). *After deleting redundant centered-zero constraints, every optimal hiddenness dual witness  $(\lambda, \alpha, \beta)$  of  $u$  satisfies*

$$\text{supp}(\lambda) \subseteq T, \quad \text{supp}(\beta) \subseteq O, \quad \text{supp}(\alpha) \subseteq Z,$$

with  $T \neq \emptyset$ , and  $O \neq \emptyset$  if and only if  $t^*(u) > 0$ .

**Registry contract** (`verbatim`).

---

Always-tight dual support: for the exposedness LP at a hidden geometrically distinct row vertex  $u$  of an exact signed idempotent  $P$  with  $\delta(P) > 0$  and nonempty visible set, every optimal hiddenness dual witness  $(\lambda, \alpha, \beta)$ , after deleting redundant centered-zero constraints, has  $\text{supp}(\lambda)$  contained in  $T$ ,  $\text{supp}(\beta)$  contained in  $O$ , and  $\text{supp}(\alpha)$  contained in  $Z$ , where  $T, O, Z$  are the rho-far, upper-box, and lower-box constraint families tight on the WHOLE primal optimal face;  $T$  is nonempty, and  $O$  is nonempty if and only if  $t^*(u) > 0$ .

---

**Proof sketch.** Pair any primal optimum  $h$  with the dual balance of Lemma 14. Using  $h(p_u) = 0$ ,  $\sum_f \lambda_f = 1$ , and  $\sum_i \beta_i = t^*(u)$  yields the *complementarity identity*

$$\sum_f \lambda_f (h(p_f) - t^*(u)) + \sum_i \alpha_i h(p_i) + \sum_i \beta_i (1 - h(p_i)) = 0. \quad (5)$$

Every summand of (5) is nonnegative by primal feasibility, so each vanishes: a positive  $\lambda_f$  forces  $h(p_f) = t^*(u)$ , a positive  $\alpha_i$  forces  $h(p_i) = 0$ , and a positive  $\beta_i$  forces  $h(p_i) = 1$ . As  $h$  ranges over all optima this places the supports in  $T$ ,  $Z$ , and  $O$ . Since  $\sum_f \lambda_f = 1$  the set  $\text{supp}(\lambda)$  is nonempty, so  $T \neq \emptyset$ ; and  $O \neq \emptyset$  iff some  $\beta_i > 0$  iff  $\sum_i \beta_i = t^*(u) > 0$  (when  $t^*(u) = 0$  the zero exposor is optimal and no upper box is tight).  $\square$

**Provenance. status:** proved, af: validated; workspace `proofs/lem-always-tight-dual-support` (export node 1), importing Lemma 14 as a validated external. Two independent fresh-codex provers produced the same complementarity core; a separate hostile verifier added the redundant-centered-zero caveat (retained in the “reduced witness” clause) and the  $t^* = 0$  edge.

**Role in the chain.** These two lemmas are the geometric engine of the hiddenness route: Lemma 14 converts “ $v$  is hidden” into a normalised transport identity on the  $\rho$ -far rows, and Lemma 15 pins that transport to the tight face. Their consumers are the depth-Markov concentration and far-row existence results of §16. Nothing here bounds the height  $H$ ; that is the open cap of the Kernel Conjecture.

## 16 Consequences of the hiddenness witness

The scales  $\tau = \sqrt{\delta}$ ,  $\rho = 4\tau$ ,  $\kappa = \frac{1}{4}\tau$ , the far set  $F_v$ , and the exposedness margin  $t^*(v)$  are those of §15. Write  $C = \text{conv}\{p_w : w \in W(P)\}$  and  $d_j = \text{dist}_1(p_j, C)$  for the depth of row  $j$ ; a hidden top vertex  $v$  has height  $H = d_v = \max_j d_j$  (**def-height**). The three results below all price the witness of Lemma 14 against an affine 1-Lipschitz functional, using the universal row-separation bound  $\|p_i - p_k\|_1 \leq (2 + 4\delta)$  of **def-signed-idempotent**.

### 16.1 Depth-Markov concentration

**Lemma 16** (**lem-hiddenness-depth-markov**). *Let  $v$  be a hidden top vertex of height  $H$  and  $(\lambda, \alpha, \beta)$  any hiddenness dual witness of  $v$  with  $\sum_i \beta_i < \kappa$ . Then for every  $c > 0$ ,*

$$\lambda\{f \in F_v : d_f > H - c\tau\} > 1 - \frac{\frac{1}{2} + \delta}{c}. \quad (6)$$

**Registry contract** (**verbatim**).

---

Hiddenness depth-Markov: for an exact signed idempotent  $P$  with  $\text{delta}(P) > 0$ , nonempty visible set  $W(P)$ , hidden top vertex  $v$  of height  $H$ , and any hiddenness dual witness  $(\text{lambda}, \text{alpha}, \text{beta})$  of  $v$  with  $\text{sum\_i } \text{beta\_i} < \text{kappa} = \text{tau}/4$  ( $\text{tau} = \text{sqrt}(\text{delta})$ ), one has for every  $c > 0$ :  $\text{lambda}\{f \text{ in } F\_v : \text{dist\_1}(p\_f, \text{conv}\{p\_w : w \text{ in } W\}) > H - c*\text{tau}\} > 1 - (1/2 + \text{delta})/c$ .

---

**Proof sketch.** Finite-dimensional  $\ell_1$  separation at  $p_v$  gives an affine functional  $\phi$  with  $\phi(p_v) = H$ , linear part of  $\ell_\infty$ -norm at most 1, and  $\phi(x) \leq \text{dist}_1(x, C)$ ; set  $\psi = H - \phi$ , so  $\psi(p_v) = 0$ . Then for every row  $i$ ,

$$0 \leq H - d_i \leq \psi(p_i) \leq \|p_v - p_i\|_1 \leq (2 + 4\delta).$$

Applying the linear part of  $\psi$  to the witness balance (3) and using  $\psi(p_v) = 0$ ,  $\alpha_i, \beta_i \geq 0$ , and  $\sum_i \beta_i < \kappa$  gives

$$\sum_f \lambda_f \psi(p_f) \leq \sum_i \beta_i \psi(p_i) < \kappa (2 + 4\delta) = \left(\frac{1}{2} + \delta\right)\tau,$$

whence  $\sum_f \lambda_f (H - d_f) < \left(\frac{1}{2} + \delta\right)\tau$ . Markov's inequality for the nonnegative random variable  $H - d_f$  under the probability  $\lambda$  on  $F_v$  yields (6).  $\square$

**Provenance.** **status:** proved, **af:** validated; workspace **proofs/lem-hiddenness-depth-markov** (export node 1), importing Lemma 14. The hostile verifier *weakened* the hypotheses: the  $c$ -parametric bound needs neither tallness nor a  $\delta$  cutoff, only  $\sum_i \beta_i < \kappa$ , and it holds for *every* witness tuple.

### 16.2 A dual certificate for the margin, and a far top-depth row

**Lemma 17** (**lem-row-far-dual-certificate**). *Let  $v$  be a geometrically distinct row vertex with  $\delta > 0$ , and set  $L_F(v) = \sum_{f: \|p_f - p_v\|_1 \geq \rho} P_{vf}^+$  with  $P_{vf}^+ = \max(P_{vf}, 0)$  and  $\nu_v$  the row- $v$  negative mass. If  $L_F(v) > 0$  then*

$$t^*(v) \leq \frac{\nu_v}{L_F(v)}. \quad (7)$$

**Registry contract** (**verbatim**).

---

Row-far dual certificate: for an exact signed idempotent  $P$  and a geometrically distinct row vertex  $v$  with  $\delta(P) > 0$  ( $\tau = \sqrt{\delta(P)}$ ,  $\rho = 4\tau$ ), writing  $L_F(v) = \sum_{f: \|p_f - p_v\|_1 \geq \rho} \max(P_{vf}, 0)$  and  $\nu_v$  the row negative mass, if  $L_F(v) > 0$  then  $t^*(v) \leq \nu_v / L_F(v)$ , where  $t^*(v)$  is the exposedness margin of def-exposed.

---

**Proof sketch.** For any admissible exposor  $h$ , row reproduction  $p_v = \sum_j P_{vj} p_j$  with  $\sum_j P_{vj} = 1$  and  $h(p_v) = 0$  give  $0 = \sum_j P_{vj} h(p_j)$ . Splitting into sign parts and using  $0 \leq h(p_j) \leq 1$  bounds the positive far contribution:

$$\sum_{f \in F_v} P_{vf}^+ h(p_f) \leq \sum_{j: P_{vj} > 0} P_{vj} h(p_j) = \sum_{j: P_{vj} < 0} (-P_{vj}) h(p_j) \leq \nu_v.$$

Dividing the weighted average by  $L_F(v) > 0$  gives  $\min_{f \in F_v} h(p_f) \leq \nu_v / L_F(v)$  for every  $h$ ; taking the supremum over  $h$  in (2) yields (7).  $\square$

**Provenance. status:** proved, af: validated; workspace proofs/lem-row-far-dual-certificate (export node 1); first-principles weak duality (no attainment needed). Sharp on the banked W29 frontier instance ( $\nu_3 / L_F = \frac{1}{81} = t^*(3)$ ).

**Lemma 18** (lem-top-slab-companion). *Let  $0 < \delta \leq \frac{1}{2}(17 - 12\sqrt{2})$ , let  $W(P)$  be nonempty, and let  $v$  be a hidden top vertex of height  $H > 13\tau$ . Then there is a row  $f$  with  $\|p_f - p_v\|_1 \geq 4\tau$  and*

$$\text{dist}_1(p_f, \text{conv}\{p_w : w \in W\}) > H - \left(\frac{1}{2} + \delta\right)\tau. \quad (8)$$

**Registry contract (verbatim).**

---

Top-slab companion: for an exact signed idempotent  $P$  with  $0 < \delta(P) \leq (17 - 12\sqrt{2})/2$ , nonempty visible set  $W(P)$ , and hidden top vertex  $v$  of height  $H > 13\tau$  ( $\tau = \sqrt{\delta(P)}$ ), there is a row  $f$  with  $\|p_f - p_v\|_1 \geq 4\tau$  and  $\text{dist}_1(p_f, \text{conv}\{p_w : w \in W\}) > H - (1/2 + \delta)\tau$ .

---

**Proof sketch.** With the same  $\psi = H - \phi$  as above ( $0 \leq \psi(p_i) \leq (2 + 4\delta)$ ), Lemma 14 supplies a witness with  $\sum_f \lambda_f \psi(p_f) \leq \sum_i \beta_i \psi(p_i) < \kappa(2 + 4\delta) = (\frac{1}{2} + \delta)\tau$ . Since the  $\lambda_f$  are a probability on  $F_v$ , some  $f \in F_v$  has  $\psi(p_f) < (\frac{1}{2} + \delta)\tau$ ; that  $f$  satisfies  $\|p_f - p_v\|_1 \geq \rho = 4\tau$  and, via  $\phi = H - \psi \leq \text{dist}_1(\cdot, C)$ , inequality (8). The constant window  $\delta \leq \frac{1}{2}(17 - 12\sqrt{2})$  makes  $13 - (\frac{1}{2} + \delta) \geq 4 + 6\sqrt{2} > 4$ , so the far row is genuinely deep.  $\square$

**Provenance. status:** proved, af: validated; workspace proofs/lem-top-slab-companion (export node 1), importing Lemma 14. The verifier checked the exact symbolic constant chain.

**Role in the chain.** Lemma 16 shows the witness mass concentrates within an  $O(\tau)$  slab below the top, Lemma 17 turns a positive far mass into a quantitative margin bound, and Lemma 18 extracts a single deep far row. Together they are the witness-side inputs to the halo-collapse height cap; the cap itself (1) stays open.

## 17 Slab and capacity primitives

The three primitives below are elementary consequences of exact idempotence. For an exact signed idempotent  $P$  (**def-signed-idempotent**) each row satisfies the *reproduction identity*

$$p_v = \sum_j P_{vj} p_j, \quad \sum_j P_{vj} = 1, \quad (9)$$

so for any affine  $h$  with  $h(p_v) = 0$  one has  $0 = \sum_j P_{vj} h(p_j)$ . Write  $P_{vj}^+ = \max(P_{vj}, 0)$ ,  $P_{vj}^- = \max(-P_{vj}, 0)$ , and  $\nu_v = \sum_j P_{vj}^-$  for the row- $v$  negative mass (**def-negative-mass**).

### 17.1 The Chebyshev low-slab pincer

**Lemma 19** (**lem-cs-low-slab-pincer**). *Let  $v$  be any row index, let  $h$  be affine with  $h(p_v) = 0$  and  $0 \leq h(p_j) \leq 1$  for every row  $j$ , and let  $s > 0$ . Then*

$$\sum_{j: h(p_j) \geq s} P_{vj}^+ \leq \frac{\nu_v}{s}. \quad (10)$$

**Registry contract** (**verbatim**).

---

CS low-slab pincer: for an exact signed idempotent  $P$ , a row index  $v$  with  $\text{nu}_v = \sum_j \max(-P_{vj}, 0)$ , an affine  $h$  with  $h(p_v) = 0$  and  $0 \leq h(p_j) \leq 1$  for every row  $j$ , and every  $s > 0$ , one has  $\sum_{\{j : h(p_j) \geq s\}} \max(P_{vj}, 0) \leq \text{nu}_v / s$ .

---

**Proof sketch.** From (9) and  $h(p_v) = 0$ , the sign split gives  $\sum_j P_{vj}^+ h(p_j) = \sum_j P_{vj}^- h(p_j) \leq \sum_j P_{vj}^- = \nu_v$ , using  $0 \leq h(p_j) \leq 1$ . On the slab  $A_s = \{j : h(p_j) \geq s\}$ ,  $s P_{vj}^+ \leq P_{vj}^+ h(p_j)$ , so  $s \sum_{A_s} P_{vj}^+ \leq \sum_j P_{vj}^+ h(p_j) \leq \nu_v$ ; divide by  $s > 0$ .  $\square$

**Provenance.** status: proved, af: validated; workspace **proofs/lem-cs-low-slab-pincer** (export node 1). The verifier confirmed the minimal hypotheses (no hiddenness, no vertex, no  $\delta > 0$ ) and that equality is attained at  $s = t^*$  on the W29 frontier instance, so (10) is sharp.

### 17.2 Row-zero capacity

**Lemma 20** (**lem-row-zero-capacity**). *Let  $i$  be a row index and  $y$  a vector with  $Py = y$ ,  $0 \leq y_j \leq 1$  for all  $j$ ,  $y_i = 0$ , and  $y_f \geq \kappa$  for all  $f$  in a set  $F$ . Then*

$$\kappa \sum_{f \in F} P_{if}^+ \leq \nu_i. \quad (11)$$

**Registry contract** (**verbatim**).

---

Row-zero capacity: for an exact signed idempotent  $P$ , a row index  $i$ , and any vector  $y$  with  $Py = y$ ,  $0 \leq y_j \leq 1$  for all  $j$ ,  $y_i = 0$ , and  $y_f \geq \kappa$  for all  $f$  in a set  $F$ , one has  $\kappa \sum_{f \in F} \max(P_{if}, 0) \leq \text{nu}_i$ , where  $\text{nu}_i$  is the row- $i$  negative mass.

---

**Proof sketch.** The  $i$ -th coordinate of  $Py = y$  with  $y_i = 0$  gives  $0 = \sum_j P_{ij} y_j$ , hence  $\sum_j P_{ij}^+ y_j = \sum_j P_{ij}^- y_j \leq \sum_j P_{ij}^- = \nu_i$  by  $0 \leq y_j \leq 1$ . On  $F$ ,  $y_f \geq \kappa$  gives  $\kappa \sum_{f \in F} P_{if}^+ \leq \sum_{f \in F} P_{if}^+ y_f \leq \sum_j P_{ij}^+ y_j \leq \nu_i$ .  $\square$

**Provenance.** status: proved, af: validated; workspace proofs/lem-row-zero-capacity (export node 1), resting on Lemma 21 in the registry (which certifies the  $Py = y$  hypothesis is the harmonic condition). The verifier made the  $y_i = 0$  hypothesis explicit.

### 17.3 The harmonic–affine bridge

**Lemma 21** (lem-harmonic-affine-bridge). *For an exact signed idempotent  $P$  with rows  $p_i = (P_{ij})_j$ , a vector  $g$  satisfies  $Pg = g$  if and only if there is a vector  $u$  with  $g_i = u \cdot p_i$  for every row index  $i$ .*

**Registry contract** (verbatim).

---

Harmonic-affine bridge: for an exact signed idempotent  $P$  with rows  $p_i = (P_{ij})_j$ , a vector  $g$  satisfies  $Pg = g$  if and only if there exists  $u$  with  $g_i = u \cdot p_i$  for every row index  $i$ ; in the forward direction  $u = g$  works ( $g_i = p_i \cdot g$ ), and the constant term of any affine representation is absorbable into  $u$  since all row sums equal 1.

---

**Proof sketch.**  $(\Rightarrow)$  If  $Pg = g$  then  $g_i = (Pg)_i = p_i \cdot g$ , so  $u = g$  works.  $(\Leftarrow)$  If  $g_i = u \cdot p_i$  then by row reproduction (9),  $(Pg)_i = \sum_j P_{ij}(u \cdot p_j) = u \cdot (\sum_j P_{ij}p_j) = u \cdot p_i = g_i$ . A constant term  $c$  in an affine representation  $g_i = c + u \cdot p_i$  is absorbed by  $u' = u + c\mathbf{1}$ , since every row sum is 1.  $\square$

**Provenance.** status: proved, af: validated; workspace proofs/lem-harmonic-affine-bridge (export node 1); two lines each direction, checked entrywise on the banked W19 rank-5 fixture.

**Role in the chain.** Lemma 21 identifies the  $P$ -harmonic vectors ( $Pg = g$ ) with affine test functions of the rows, so an admissible expositor is exactly a box-constrained harmonic vector; Lemmas 19 and 20 are then the two capacity estimates that price positive far mass against  $\nu_v$ . They feed the same halo-collapse and starvation arguments as the witness lemmas of §16.

## 18 Parametric and depth- $d$ halo collapse

Both results generalise the af-validated halo-collapse bridge 6 by making the halo width a free parameter. Fix an exact signed idempotent  $P$  with  $0 < \delta \leq \frac{1}{4}$  and nonempty visible set  $W(P)$ ; write  $C = \text{conv}\{p_w : w \in W\}$ ,  $d_j = \text{dist}_1(p_j, C)$ ,  $\tau = \sqrt{\delta}$ , and  $\alpha_j = P_{vj}$  with parts  $\alpha_j^+, \alpha_j^-$  and  $\nu_v = \sum_j \alpha_j^-$ . For a halo width  $a > 0$  the mass placed strictly outside the  $a\tau$ -halo of  $C$  is  $\sigma_a = \sum_{j: d_j > a\tau} \alpha_j^+$ , and  $\sigma = \sum_{j: d_j > 0} \alpha_j^+$  is the invisible mass (`def-invisible-mass`).

### 18.1 The parametric bridge

**Lemma 22** (`lem-parametric-halo-collapse`). *Let  $v$  be a hidden top vertex of height  $H$  and  $a > 0$ . Then*

$$H(1 - \sigma_a) \leq (\sigma - \sigma_a)a\tau + \nu_v(2 + 4\delta). \quad (12)$$

**Registry contract** (`verbatim`).

---

Parametric halo collapse: for an exact signed idempotent P with  $0 < \text{delta}(P) \leq 1/4$ , nonempty visible set  $W(P)$ , hidden top vertex  $v$  of height  $H$ , and any halo width  $a > 0$ , writing `sigma_a` for the positive coefficient mass  $v$  places on rows at  $\text{ell-1}$  distance  $> a\tau$  from  $\text{conv } W$  ( $\tau = \sqrt{\text{delta}}$ ), `sigma` for the invisible mass, and `nu_v` for the row negative mass, one has  $H*(1 - \text{sigma}_a) \leq (\text{sigma} - \text{sigma}_a)*a\tau + \text{nu}_v*(2 + 4*\text{delta})$ .

---

**Proof sketch.** If  $\sigma_a \geq 1$  the left side is  $\leq 0$  and the right side is  $\geq 0$ . Otherwise put  $A = \{j : d_j > a\tau\}$  and

$$q = \frac{1}{1 - \sigma_a} \left( \sum_{j \notin A} \alpha_j^+ p_j - \sum_k \alpha_k^- p_k \right),$$

so that by row reproduction and mass split ( $\sum_j \alpha_j^+ = 1 + \nu_v$ , 3)  $p_v = \sum_{j \in A} \alpha_j^+ p_j + (1 - \sigma_a)q$  is a genuine convex combination. The lower residual estimate 4 gives  $H \leq \text{dist}_1(q, C)$ , while the upper residual estimate 5 with  $D_k = (2 + 4\delta)$  gives  $(1 - \sigma_a)\text{dist}_1(q, C) \leq \sum_{j \notin A} \alpha_j^+ d_j + \nu_v(2 + 4\delta)$ . Pricing the non-halo rows by  $\sum_{j \notin A} \alpha_j^+ d_j \leq (\sigma - \sigma_a)a\tau$  and multiplying the first bound by  $1 - \sigma_a > 0$  yields (12).  $\square$

**Provenance.** `status:` proved, `af:` validated; `workspace proofs/lem-parametric-halo-collapse` (export node 1), importing 3, 4, 5. At  $a = \frac{1}{4}$  it recovers the 6 contract `verbatim`; the verifier checked that calibration and the rank-5 fixture.

### 18.2 The depth- $d$ version

**Lemma 23** (`lem-depth-d-halo-collapse`). *Let  $v$  be any row index with depth  $d = \text{dist}_1(p_v, C)$  and let  $a > 0$ . With  $\sigma_a(v) = \sum_{d_j > a\tau} P_{vj}^+$ ,  $\sigma(v) = \sum_{d_j > 0} P_{vj}^+$ , and the deeper-row correction  $C_a(v) = \sum_{d_j > a\tau, d_j > d} P_{vj}^+(d_j - d)$ ,*

$$d(1 - \sigma_a(v)) \leq (\sigma(v) - \sigma_a(v))a\tau + \nu_v(2 + 4\delta) + C_a(v). \quad (13)$$

**Registry contract** (`verbatim`).

---

Depth-d halo collapse: for an exact signed idempotent P with  $0 < \text{delta}(P) \leq 1/4$  and nonempty visible set  $W(P)$ , any row index  $v$  with  $d = \text{dist}_1(p_v, \text{conv}\{p_w : w \in W\})$ , and any halo width  $a > 0$ , writing  $d_j = \text{dist}_1(p_j, \text{conv } W)$ ,  $\text{sigma}_a(v) = \text{sum over } \{j : d_j > a\tau\} \text{ of } \max(P_{vj}, 0)$  ( $\tau = \sqrt{\text{delta}}$ ),  $\text{sigma}(v) = \text{sum over } \{j : d_j > 0\} \text{ of } \max(P_{vj}, 0)$ , and the deeper-row correction  $C_a(v) = \text{sum over } \{j : d_j > a\tau, d_j > d\} \text{ of } \max(P_{vj}, 0) * (d_j - d)$ , one has  $d*(1 - \text{sigma}_a(v)) \leq (\text{sigma}(v) - \text{sigma}_a(v))*a\tau + \text{nu}_v*(2 + 4*\text{delta}) + C_a(v)$ .

---

= sqrt(delta)), sigma(v) = sum over {j : d\_j > 0} of max(P\_vj, 0), nu\_v the row negative mass, and C\_a(v) = sum over {j : d\_j > a\*tau, d\_j > d} of max(P\_vj, 0)\*(d\_j - d), one has d\*(1 - sigma\_a(v)) <= (sigma(v) - sigma\_a(v))\*a\*tau + nu\_v\*(2 + 4\*delta) + C\_a(v).

---

**Proof sketch.** The trivial branch  $\sigma_a(v) \geq 1$  is as before. Otherwise the same residual  $q$  (now with  $B = \{j : d_j \leq a\tau\}$ ) gives  $p_v = \sum_{j \in A} P_{vj}^+ p_j + (1 - \sigma_a)q$ . Convexity of  $x \mapsto \text{dist}_1(x, C)$  gives  $d \leq \sum_{j \in A} P_{vj}^+ d_j + (1 - \sigma_a) \text{dist}_1(q, C)$ ; splitting  $A$  at  $d_j \leq d$  versus  $d_j > d$  replaces  $\sum_{j \in A} P_{vj}^+ d_j$  by  $\sigma_a(v)d + C_a(v)$ , so  $(1 - \sigma_a)d \leq C_a(v) + (1 - \sigma_a) \text{dist}_1(q, C)$ . Pricing  $\text{dist}_1(q, C)$  through 5 exactly as in the parametric case gives (13). At a hidden top ( $d = H$ ,  $C_a = 0$ ) it recovers (12).  $\square$

**Provenance. status:** proved, **af:** validated; workspace `proofs/lem-depth-d-halo-collapse` (export node 1), importing 3, 5. The verifier legitimised the convex split and checked the top-row calibration against 22.

**Role in the chain.** These are the halo-collapse bounds at arbitrary width and depth: the free parameter  $a$  is what lets the genuine-mass bootstrap of §19 choose a halo that separates hidden from visible rows. They tighten but do not close the height cap of the Kernel Conjecture 1.



## 19 Genuine-mass disintegration and top concentration

Keep the notation of §18:  $C = \text{conv}\{p_w : w \in W(P)\}$ ,  $d_j = \text{dist}_1(p_j, C)$ ,  $\tau = \sqrt{\delta}$ , height  $H$ , and  $P_{ij}^+ = \max(P_{ij}, 0)$ . Fix a halo width  $a > 0$  and the genuine (deep-halo) index set  $G_a = \{j : d_j > a\tau\}$ , assumed nonempty. The two results below are the “bootstrap” step: they show that the mass a row sends into  $G_a$  is either supported on genuinely hidden vertices or is small.

### 19.1 Disintegration of the halo mass

**Lemma 24** (lem-genuine-disintegration). *Fix for every row  $j$  a convex vertex representation  $p_j = \sum_v \lambda_{jv} p_v$  over geometrically distinct row vertices, and let  $g = P\mathbf{1}_{G_a}$ . Then every row index  $i$  satisfies*

$$g_i \leq M_i^a + \sum_{j \in G_a} P_{ij}^+ \frac{H - d_j}{H - a\tau}, \quad M_i^a = \sum_{j \in G_a} P_{ij}^+ \sum_{v: d_v > a\tau} \lambda_{jv}, \quad (14)$$

where  $M_i^a \geq 0$  is positive mass supported entirely on hidden row vertices of depth in  $(a\tau, H]$ .

**Registry contract** (verbatim).

---

Genuine-mass disintegration: for an exact signed idempotent  $P$  with  $0 < \delta(P) \leq 1/4$ , nonempty visible set  $W(P)$ , height  $H$ , halo width  $a > 0$  with  $G_a = \{j : \text{dist}_1(p_j, \text{conv } W) > a\tau\}$  nonempty ( $\tau = \sqrt{\delta(P)}$ ), fix for every row  $j$  a vertex representation  $p_j = \sum_v \lambda_{jv} p_v$  over geometrically distinct row vertices; then every row index  $i$  satisfies  $g_i \leq M_i^a + \sum_{j \in G_a} P_{ij}^+ \frac{H - d_j}{H - a\tau}$ , where  $g = P\mathbf{1}_{G_a}$ ,  $d_j = \text{dist}_1(p_j, \text{conv } W)$ ,  $P_{ij}^+ = \max(P_{ij}, 0)$ , and  $M_i^a = \sum_{j \in G_a} P_{ij}^+ \sum_{v: d_v > a\tau} \lambda_{jv}$  is positive mass supported entirely on HIDDEN row vertices at depth in  $(a\tau, H]$ .

---

**Proof sketch.** Nonemptiness of  $G_a$  forces  $H > a\tau$ . Applying the upper residual estimate 5 to the vertex representation gives the depth-convexity bound  $d_j \leq \sum_v \lambda_{jv} d_v$ . Splitting each  $j \in G_a$  into shallow mass  $L_j = \sum_{d_v \leq a\tau} \lambda_{jv}$  and deep mass  $U_j = 1 - L_j$  and using  $d_v \leq a\tau$  on the first class,  $d_v \leq H$  on the second, yields  $d_j \leq a\tau L_j + H U_j$ , hence  $L_j \leq (H - d_j)/(H - a\tau)$ . Since  $g_i = \sum_{j \in G_a} P_{ij} \leq \sum_{j \in G_a} P_{ij}^+$  and  $1 = L_j + U_j$ , weighting the  $L_j$ -bound by  $P_{ij}^+ \geq 0$  gives (14); the  $U_j$ -part is  $M_i^a$ , whose vertices have  $d_v > a\tau$  and hence are hidden of depth  $(a\tau, H]$ .  $\square$

**Provenance.** status: proved, af: validated; workspace proofs/lem-genuine-disintegration (export node 1), importing 5. The verifier checked the classification, extreme-point identification, and denominators on the banked rank-3 and rank-5 instances (step 3 of sketch M1).

### 19.2 Top concentration

**Lemma 25** (lem-top-concentration). *Let  $v$  be a hidden top vertex of height  $H$  and  $a > 0$  with  $H > a\tau$ . The positive mass  $v$  places outside the genuine set  $G_a$  satisfies*

$$\sum_{j \notin G_a} P_{vj}^+ \leq \frac{\nu_v (2 + 4\delta)}{H - a\tau}. \quad (15)$$

**Registry contract** (verbatim).

---

Top concentration: for an exact signed idempotent  $P$  with  $0 < \delta(P) \leq 1/4$ , nonempty visible set  $W(P)$ , hidden top vertex  $v$  of height  $H$ , and halo width  $a > 0$  with  $H > a\tau$  ( $\tau = \sqrt{\delta(P)}$ ), the positive mass  $v$  places outside the genuine set satisfies  $\sum_{j \notin G_a} P_{vj}^+ \leq \frac{\nu_v (2 + 4\delta)}{H - a\tau}$ .

---

$\text{nu\_v} \cdot (2 + 4\delta) / (H - a\tau)$ , where  $G_a = \{j : \text{dist}_1(p_j, \text{conv}\{p_w : w \in W\}) > a\tau\}$  and  $\text{nu\_v}$  is the row- $v$  negative mass.

---

**Proof sketch.** Choose an  $\ell_1/\ell_\infty$  support functional  $\phi(x) = u \cdot x + b$  with  $\|u\|_\infty \leq 1$ ,  $\phi \leq 0$  on  $C$ , and  $\phi(p_v) = H$ ; set  $t_j = H - \phi(p_j)$ . Then  $0 \leq t_j \leq (2 + 4\delta)$  (row separation),  $t_j \geq H - d_j \geq H - a\tau$  for  $j \notin G_a$ , and by row reproduction  $\sum_j P_{vj} t_j = 0$ . The sign split gives  $\sum_j P_{vj}^+ t_j = \sum_j P_{vj}^- t_j \leq (2 + 4\delta) \sum_j P_{vj}^- = \nu_v(2 + 4\delta)$ . Restricting to  $j \notin G_a$ ,  $(H - a\tau) \sum_{j \notin G_a} P_{vj}^+ \leq \sum_j P_{vj}^+ t_j \leq \nu_v(2 + 4\delta)$ ; divide by  $H - a\tau > 0$ .  $\square$

**Provenance.** status: proved, af: validated; workspace proofs/lem-top-concentration (export node 1); first-principles support functional, no imports. The verifier ran an exact LP support-functional test on the banked rank-5 instance (step 4 decider).

**Role in the chain.** Lemma 25 shows a hidden top vertex sends all but an  $O(\nu_v/H)$  fraction of its positive mass into the genuine set  $G_a$ , and Lemma 24 then re-expresses that genuine mass as mass on strictly hidden vertices plus a controlled shallow remainder. Iterating these against the depth- $d$  collapse 23 is the  $g$ -bootstrap intended to force  $H = O(\tau)$ ; the closing cap remains the open Kernel Conjecture 1.

## 20 The bounded-slab starvation completion obstruction

This standalone obstruction rules out a rank-three completion of the exchange-starvation configuration without any auxiliary constant  $K$ . Fix a finite index set  $I$ , a real  $A \in [4, 6]$ , a real  $\tau \in (0, \frac{1}{256}]$ , and set

$$t = \tau^2, \quad a = \frac{\tau}{1 + \tau}.$$

**Lemma 26** (`lem-starvation-completion-obstruction`). *There is no rank-three exact signed idempotent  $P$  (so  $P^2 = P$ ,  $P\mathbf{1} = \mathbf{1}$ ,  $\text{rank } P = 3$ ) with row negative mass  $\nu_i \leq t$  for every  $i \in I$  that carries five distinct full row-point fibers, represented by  $v, w, f, z, o$ , such that, writing  $D = p_z - p_v$  and  $E = p_o - p_v$  (linearly independent),*

$$\|D\|_1 = \tau, \quad p_f - p_v = -AD + tE, \quad p_w - p_v = a(p_f - p_v),$$

*with top-row fiber masses  $c_v = 1 - \tau$ ,  $c_w = \tau + t$ ,  $c_f = -t$ , and  $c_Q = 0$  for every other fiber  $Q$ , where every full row-point fiber has unique coordinates  $p_Q = p_v + x_Q D + y_Q E$  and every nonactor support fiber satisfies  $p_Q \in \text{conv}\{p_v, p_w, p_f, p_z, p_o\}$  or  $0 \leq y_Q \leq 1$ .*

**Registry contract** (`verbatim`).

---

Bounded-slab starvation completion obstruction (K-free): for every finite  $I$ , every real  $A$  in  $[4, 6]$ , every real  $\tau$  in  $(0, 1/256]$ , and  $t := \tau^2$ ,  $a := \tau/(1+\tau)$ , there is no rank-three exact signed idempotent  $P$  ( $P^2 = P$ ,  $P\mathbf{1} = \mathbf{1}$ ,  $\text{rank } P = 3$ ) with row negative mass  $\nu_i \leq t$  for every  $i$  in  $I$  having five distinct full row-point fibers represented by  $v, w, f, z, o$  such that, with  $D := p_z - p_v$  and  $E := p_o - p_v$  linearly independent,  $\|D\|_1 = \tau$ ,  $p_f - p_v = -A*D + t*E$ ,  $p_w - p_v = a*(p_f - p_v)$ , top-row fiber masses  $c_v = 1 - \tau$ ,  $c_w = \tau + t$ ,  $c_f = -t$ , and  $c_Q = 0$  for every other full row-point fiber  $Q$ , every full row-point fiber  $Q$  has unique reals  $x_Q, y_Q$  with  $p_Q = p_v + x_Q*D + y_Q*E$ , and every nonactor support fiber  $Q$  satisfies either  $p_Q$  in  $\text{conv}\{p_v, p_w, p_f, p_z, p_o\}$  or  $0 \leq y_Q \leq 1$ .

---

**Proof sketch (by contradiction).** Suppose such a  $P$  exists. Group the coordinate indices into fibers  $Q$  (classes of equal rows) and set  $c_Q = \sum_{j \in Q} P_{vj}$ ,  $d_Q = \sum_{j \in Q} (P_{zj} - P_{vj})$ ,  $e_Q = \sum_{j \in Q} (P_{oj} - P_{vj})$ . The negative-mass hypothesis  $\nu_i \leq t$  gives the subset ledger  $-t \leq P_i(S) \leq 1 + t$  and  $\|p_i\|_1 \leq 1 + 2t$  for every row  $i$  and subset  $S$ . Subtracting the  $v$ -row from the  $z$ -row of  $P^2 = P$  gives  $DP = D$  with  $\sum_j D_j = 0$ ; expanding  $D = \sum_j D_j p_j$  in the coordinates  $p_j = p_v + x_j D + y_j E$  and using linear independence of  $D, E$  produces the *exact unit moment*

$$\sum_Q x_Q d_Q = 1. \tag{16}$$

Every actor and interior fiber has bounded  $x$ -coordinate,  $|x_Q| \leq A$ , so the hull part contributes at most  $A\tau$  (since  $\sum_{Q \in \text{Hin}} |d_Q| \leq \|D\|_1 = \tau$ ). For an exterior fiber  $Q$  the disjunction forces  $0 \leq y_Q \leq 1$ , and the coordinate identity with the row-norm bounds gives the lever estimate  $|x_Q|\tau \leq 2 + 4t$ ; a single global budget from the row- $z$ , row- $f$ , and row- $o$  subset bounds gives  $\sum_{Q \in \text{Ext}} |d_Q| \leq t(1 + (2 + t)/A)$ , with *no* dependence on the number of fibers. Combining,

$$1 = \sum_Q x_Q d_Q \leq \tau \left[ A + (2 + 4t) \left( 1 + \frac{2+t}{A} \right) \right].$$

But  $0 < \tau \leq \frac{1}{256}$  gives  $t < \frac{1}{4}$ ,  $2 + 4t < 3$ ,  $(2 + t)/A \leq 3/4$ , and  $A \leq 6$ , so the right side is strictly below  $\frac{1}{256} \left[ 6 + 3 \left( 1 + \frac{3}{4} \right) \right] = \frac{45}{1024} < 1$  — contradicting (16).  $\square$

**Provenance.** status: proved, af: validated; workspace proofs/lem-starvation-completion-obstruction (export node 1). W59 wave: a codex prover paper proof from first principles (guided by, but independent of, the W57/W58 exact Farkas certificates); a fresh hostile codex verifier returned VALID-WITH-CORRECTIONS (a single index-level notation correction applied). Reviewer  $\neq$  author (owner: B).

**Role in the chain.** This is the “K-free” closure of the exchange-starvation leaf: the earlier reductions of that branch depended on an auxiliary completion constant, and this obstruction removes it by showing the rank-three completion cannot exist in the bounded-slab regime. It is the verified four-leaf-system input recorded in the live proof sketch; it does not by itself settle op-classical, which remains open.

## 21 The H-X financing engine

The three results of this section form the demand side of the H-X front: an exact signed idempotent forces two separated points of its row polytope to finance positive coefficient mass on every high-lever fiber set. All quantities are full-fiber (clone-invariant) and the picture is signed throughout. For a finite exact signed idempotent  $P$  write  $K(P) = \text{conv}\{p_i\}$  for the row polytope, group coordinate indices into full row-point fibers  $Q$  (classes of equal rows), and for a point  $r \in \mathbb{R}^I$  set  $r^+(S) = \sum_{Q \in S} \sum_{j \in Q} \max(r_j, 0)$  and  $\nu(r) = \sum_j \max(-r_j, 0)$ .

### 21.1 The basis-free unit transverse moment

**Lemma 27** (`lem-hx-transverse-moment-identity`). *For all points  $q_0 \neq q_1$  of  $K(P)$  and every affine  $\chi$  on  $\mathbb{R}^I$  with  $\chi(q_0) = 0$  and  $\chi(q_1) = 1$ , the full row-point fibers satisfy  $\sum_Q d_Q \chi(p_Q) = 1$ , where  $d_Q = \sum_{j \in Q} (q_{1j} - q_{0j})$ .*

**Registry contract** (`verbatim`).

---

For every finite exact signed idempotent P, all points  $q_0 \neq q_1$  of the row polytope  $K(P) = \text{conv}\{p_i\}$ , and every affine function  $\chi$  on  $\mathbb{R}^I$  with  $\chi(q_0) = 0$  and  $\chi(q_1) = 1$ , the full row-point fibers satisfy  $\sum_Q d_Q \chi(p_Q) = 1$ , where  $d_Q = \sum_{j \in Q} (q_{1j} - q_{0j})$ .

---

**Proof sketch.** Every point of  $K(P)$  is a convex combination of rows of  $P$  and hence fixed by right multiplication:  $q_i P = q_i$ . Subtracting,  $D := q_1 - q_0$  is left-fixed,  $D = \sum_j D_j p_j$ , with  $\sum_j D_j = 0$  (both points have coordinate sum one). Applying the linear part  $L$  of  $\chi$  and using the endpoint normalization,  $1 = L(D) = \sum_j D_j L(p_j) = \sum_j D_j \chi(p_j)$  (the affine constant dies against  $\sum_j D_j = 0$ ); grouping the finite sum by fibers ( $\chi(p_Q)$  constant on a fiber) gives the display. No rank hypothesis, no basis, no coordinate chart.  $\square$

### 21.2 The two-sign-union budget converter

**Lemma 28** (`lem-hx-signed-variation-ledger`). *For every ordered pair  $(a, b)$  of points of  $K(P)$  and every set  $S$  of full row-point fibers,  $\sum_{Q \in S} |d_Q| \leq a^+(S) + b^+(S) + \nu(a) + \nu(b)$ , where  $d_Q = \sum_{j \in Q} (a_j - b_j)$ .*

**Registry contract** (`verbatim`).

---

For every finite exact signed idempotent P, every ordered pair  $(a, b)$  of points of the row polytope  $K(P)$ , and every set  $S$  of full row-point fibers,  $\sum_{Q \in S} |d_Q| \leq a^+(S) + b^+(S) + \nu(a) + \nu(b)$ , where  $d_Q = \sum_{j \in Q} (a_j - b_j)$ ,  $a^+(S) = \sum_{Q \in S} \sum_{j \in Q} \max(a_j, 0)$ , and  $\nu(a) = \sum_j \max(-a_j, 0)$ .

---

**Proof sketch.** Partition  $S$  by the sign of the aggregate  $d_Q$  into  $S_+$  and  $S_-$  (fibers with  $d_Q = 0$  contribute nothing). On the genuine index subset underlying  $S_+$ ,  $\sum_{S_+} d_Q = a(S_+) - b(S_+) \leq a^+(S_+) + \nu(b)$ , since  $a(S_+) \leq a^+(S_+)$  and  $-b(S_+) \leq \nu(b)$  by the subset ledger of a mass-one point with negative part  $\nu(b)$ ; symmetrically on  $S_-$ . Each budget is paid once per sign union, never per fiber — the  $K$ -free discipline of the starvation proof (§20), scalarized.  $\square$

### 21.3 The financing floor

**Lemma 29** (`lem-hx-financing-floor`). *For every ordered pair  $(a, b)$  of points of  $K(P)$  with  $a \neq b$ , every affine  $\chi$  with  $\chi(a) - \chi(b) = 1$ , all reals  $A > 0$  and  $\Lambda > 0$ , and every set  $N$  of full row-point fibers with  $|\chi(p_Q)| \leq A$  on  $N$  and  $|\chi(p_Q)| \leq \Lambda$  off  $N$ , the complement  $F$  of  $N$  satisfies*

$$a^+(F) + b^+(F) \geq \frac{1 - A\ell_\chi}{\Lambda} - \nu(a) - \nu(b), \quad \ell_\chi := \sum_Q |d_Q|.$$

**Registry contract** (`verbatim`).

---

For every finite exact signed idempotent  $P$ , every ordered pair  $(a, b)$  of points of the row polytope  $K(P)$  with  $a \neq b$ , every affine  $\chi$  with  $\chi(a) - \chi(b) = 1$ , all reals  $A > 0$  and  $\Lambda > 0$ , and every set  $N$  of full row-point fibers with  $|\chi(p_Q)| \leq A$  for every  $Q$  in  $N$  and  $|\chi(p_Q)| \leq \Lambda$  for every  $Q$  not in  $N$ , the complement  $F$  of  $N$  satisfies  $a^+(F) + b^+(F) \geq (1 - A \cdot l_\chi) / \Lambda - \nu(a) - \nu(b)$ , where  $l_\chi = \sum_Q |d_Q|$  and  $d_Q = \sum_{j \in Q} (a_j - b_j)$ .

---

**Proof sketch.** Recentring  $\chi$  at  $b$  reduces to the endpoint normalization of Lemma 27, whose unit moment splits over  $N \cup F$  as  $1 \leq A \sum_N |d_Q| + \Lambda \sum_F |d_Q| \leq A \ell_\chi + \Lambda (a^+(F) + b^+(F) + \nu(a) + \nu(b))$ , using the lever bounds on each part and Lemma 28 with  $S = F$ ; rearrange. In the recentred instantiation ( $\chi = \psi_c$  the recentred sign functional of the pair,  $N$  a metric ball around any  $c \in K(P)$ ,  $\Lambda = (2+4\delta)/\|a-b\|_1$  via the signed row-diameter bound) the floor reads: *two points at  $\ell^1$ -separation  $\ell$  must jointly ship  $\gtrsim (1 - A\ell_\chi)\ell/(2+4\delta)$  positive mass outside every low-lever ball* — the rank-free, slab-free replacement for the W59 lever geometry.  $\square$

**Quantifier correction (W61).** The W60 contract quantified over all reals  $A$ ; with  $N = \emptyset$  that hypothesis is vacuous and the  $A < 0$  reading is false (already the  $2 \times 2$  identity idempotent gives a counterexample). The af elevation surfaced this (challenge `ch-9388e571`); the contract now states the  $A > 0$  form the W60 proof establishes. The retraction of the earlier reading is recorded in `docs/LEARNINGS.md`; all registry consumers instantiate  $A > 0$ .

**Provenance.** All three: `status: proved`, `af: validated`, W60 wave codex prover + fresh hostile batched codex verifier (`VERDICT-W60-ENGINE.md`), then W61 per-node af orchestration — moment identity 14/14 nodes, ledger 11/11, floor 12 validated + 2 archived, all taint clean; workspaces `proofs/lem-hx-transverse-moment-identity`, `proofs/lem-hx-signed-variation-ledger`, `proofs/lem-hx-f` (export node 1 each). Reviewer  $\neq$  author throughout (`owner: B`).

**Role in the chain.** This engine is the quantitative demand side consumed by the W60 route fork on the H-X cell (`conj-s11a-off-diagonal-cell`) and by the L6.5 co-top coupling front; the starvation generalization and the long-range coupling instantiation are typeset in §22. It supplies lower bounds only; it does not by itself settle any open cell, and `op-classical` remains open.

## 22 Robust scalar starvation and forced exterior coupling

This section typesets the two remaining engine results: the generalization of the starvation obstruction of §20 — rank-free, slab-free, with the exact tableau pins relaxed to a metric window and the zero-top hypothesis to an  $O(\delta)$  tail cap — and the ball instantiation of the financing floor of §21. Notation as there; additionally  $\delta = \delta(P)$ ,  $\tau = \sqrt{\delta}$ , and  $c_Q = \sum_{j \in Q} P_{vj}$  for a distinguished fiber representative  $v$ .

### 22.1 Robust scalar starvation

**Lemma 30** (`lem-hx-robust-scalar-starvation`). *For all reals  $K_R, L, K_C \geq 0$  there is a universal  $\delta_R(K_R, L, K_C) \in (0, 2^{-16}]$  such that no finite exact signed idempotent  $P$  with  $0 < \delta \leq \delta_R$  admits full row-point fibers represented by  $v$  and  $f$ , a real  $A \geq 4$  and a point  $q \in K(P)$  with  $\tau/2 \leq \|q - p_v\|_1 \leq 2\tau$  and  $\|p_f - p_v + A(q - p_v)\|_1 \leq K_R\delta$ , and an affine  $\chi$  with  $\chi(p_v) = 0$ ,  $\chi(q) = 1$ , and  $|\chi(x) - \chi(y)| \leq \|x - y\|_1 / \|q - p_v\|_1$  on  $K(P)$ , such that  $\sum_{Q: |\chi(p_Q)| > L} \max(c_Q, 0) \leq K_C \delta$ .*

**Registry contract (verbatim).**

---

For all reals  $K_R, L, K_C \geq 0$  there is a universal  $\delta_R(K_R, L, K_C)$  in  $(0, 2^{-16}]$  such that no finite exact signed idempotent  $P$  with  $0 < \delta(P) \leq \delta_R$  admits full row-point fibers represented by  $v$  and  $f$ , a real  $A \geq 4$  and a point  $q$  of the row polytope  $K(P)$  with  $\sqrt{\delta(P)}/2 \leq \|q - p_v\|_1 \leq 2\sqrt{\delta(P)}$  and  $\|p_f - p_v + A(q - p_v)\|_1 \leq K_R\delta(P)$ , and an affine  $\chi$  with  $\chi(p_v) = 0$ ,  $\chi(q) = 1$ , and  $|\chi(x) - \chi(y)| \leq \|x - y\|_1 / \|q - p_v\|_1$  for all  $x, y$  in  $K(P)$ , such that  $\sum_{Q: |\chi(p_Q)| > L} \max(c_Q, 0) \leq K_C\delta(P)$ , where  $c_Q = \sum_{j \in Q} P_{vj}$ .

---

**Explicit ceiling.** With  $B = K_C + 1 + (K_C + K_R + 1)/4$  and  $H = 2L + 6B$ :  $\delta_R = \min\{2^{-16}, (4H^2)^{-1}\}$ .

**Proof sketch (by contradiction).** Set  $D = q - p_v$  and apply the unit moment (Lemma 27 at the pair  $(p_v, q)$ ):  $\sum_Q d_Q \chi(p_Q) = 1$ . The core ( $|\chi(p_Q)| \leq L$ ) costs at most  $L\|D\|_1 \leq 2L\tau$  since grouping decreases  $\ell^1$  variation. On the tail, split by the sign of  $d_Q$ : the negative union is paid by the positive  $v$ -row top tail plus  $\nu_v$ , at most  $(K_C + 1)\delta$ ; the positive union is routed through the finance-row identity  $p_f = p_v - AD + r$ ,  $\|r\|_1 \leq K_R\delta$ , whose subset budget gives at most  $(K_C + K_R + 1)\delta/A \leq (K_C + K_R + 1)\delta/4$ . The intrinsic Lipschitz bound caps the tail lever by the row diameter over  $\|D\|_1$ , so the tail cost is  $O(\delta/\tau) = O(\tau)$ ; below  $\delta_R$  the total is  $< 1$ , contradicting the moment.  $\square$

**Calibration and sharpness of the hypotheses.** (i) The starvation obstruction of §20 sits at the fixed calibration  $(K_R, L, K_C) = (3, 1, 0)$  with  $\delta_R(3, 1, 0) = 2^{-16}$ , slightly roomier than the W59 close. (ii) The af verifier's independent rank-two exact near-example satisfies every hypothesis *except* the tail cap ( $\text{Tail}_1 \approx 1$ ) — the top-tail cap is the indispensable obstruction, not a consequence of the scaffold. The lemma supplies neither the actor pair  $(A, q)$  nor the tail cap from an H-X datum; producing those is exactly the open hard residual (the W60 route fork).

### 22.2 Forced exterior coupling

**Lemma 31** (`lem-hx-forced-exterior-coupling`). *For every pair of row indices  $(r, s)$  and every point  $c \in K(P)$ , the full row-point fibers  $Q$  with  $\|p_Q - c\|_1 > 1/2$  jointly carry positive coefficient mass*

$$P_r^+ + P_s^+ \geq \frac{\|p_r - p_s\|_1}{2(2 + 4\delta)} - 2\delta.$$

## Registry contract (verbatim).

---

For every finite exact signed idempotent  $P$ , every pair of row indices  $(r,s)$ , and every point  $c$  of the row polytope  $K(P)$ , the full row-point fibers  $Q$  with  $\|p_Q - c\|_1 > 1/2$  jointly carry positive coefficient mass  $P_r^+ + P_s^+$  at least  $\|p_r - p_s\|_1 / (2 \cdot (2 + 4 \cdot \delta(P))) - 2 \cdot \delta(P)$ .

---

**Proof sketch.** Instantiate the financing floor (Lemma 29) at the pair  $(p_r, p_s)$  with the recentred sign functional  $\psi_c$ , the ball  $N = \{Q : \|p_Q - c\|_1 \leq 1/2\}$ ,  $A \ell_\chi = 1/2$ , and  $\Lambda = (2 + 4\delta)/\|p_r - p_s\|_1$  from the signed row-diameter bound  $\|p_i\|_1 \leq 1 + 2\nu_i$ ; the floor rearranges to the display. At  $\delta = 0$  this is the quantitative shadow of block disjointness: two mixture rows at separation  $\ell$  ship  $\Omega(\ell)$  mass to  $\ell^1$ -spread recurrent fibers. Exact vacuity threshold:  $\|p_r - p_s\|_1 \leq 8\delta + 16\delta^2$ .  $\square$

**Provenance.** Both: **status:** `proved`, **af:** `validated`, W60 wave codex prover + fresh hostile batched codex verifier, then W61 per-node af orchestration — starvation 12/12 nodes, coupling 12/12, taint clean; workspaces `proofs/lem-hx-robust-scalar-starvation`, `proofs/lem-hx-forced-exterior-` (export node 1 each). Reviewer  $\neq$  author throughout (**owner:** `B`).

**Role in the chain.** Lemma 30 retires the rank and slab restrictions of §20 at the mechanism level and de-pins the tableau metric to the window  $[\tau/2, 2\tau]$ ; Lemma 31 is the first forced long-range positive-financing lower bound (feeding the L6.5 coupling front). Neither settles any open cell; `op-classical` remains open.



## 23 Status ledger for non-validated registry results

The rows below are anchors, not upgrades. They give every remaining registry result a report label while preserving the registry status. Of the twenty-nine **af**-validated registry results, twenty-six are reproduced as their own sections (enumerated in the roadmap, §1) and are not duplicated in this ledger. The three validated results not yet given a dedicated section (**lem-collateral-import**, **lem-cross-pivot-cancellation**, **lem-import-reduction**) are anchored below at their true **af**-validated status.

Table 1: Registry status ledger for report anchors outside the thirteen validated sections.

| Registry id                      | Status                   | Contract gloss                                                                                                                                                 |
|----------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>conj-extcb</b>                | conjecture               | EXT-CB amplified extension gap: one level-one unitary and one extension map must carry uniform estimates at every amplification.                               |
| <b>conj-hcb</b>                  | conjecture               | H-CB complete column-Hilbert and Ha-module estimates, uniform in amplification and all dimensions; open.                                                       |
| <b>conj-degenerate-payment</b>   | conjecture               | Pivot-local weighted near-degenerate payment bound; the payment horn of the (EX) argmin charge, exact-tested on the wave zoo but unproved.                     |
| <b>conj-degenerate-transport</b> | conjecture               | Degenerate transverse transport bound; the remaining payment-horn gap after the wave-12 lift, exact-tested but unproved.                                       |
| <b>conj-ex</b>                   | conjecture               | (EX) chart-bound working form for exact signed idempotents; still a conjectural input equivalent to <b>conj-kernel</b> in the registry.                        |
| <b>conj-kernel</b>               | conjecture               | Kernel Conjecture, the geometric input intended to close <b>op-exposed-hull</b> and hence <b>op-classical</b> .                                                |
| <b>conj-no-free-frontier</b>     | conjecture               | Exposedness-absorption statement from the frontier analysis; retained as conjectural and not elevated.                                                         |
| <b>conj-rh</b>                   | conjecture               | Repaired orphan horn: post-fan orphan demand bounded by the unified class/signed/own-negativity budget plus fan residual; exact floor four on the coefficient. |
| <b>conj-sc</b>                   | conjecture               | Argmin self-support/cancellation control for non-fan beta-positive rows; reduced to the pivot-removing blocker theorem, collateral branch open.                |
| <b>conj-skinny-shadow-cap</b>    | conjecture               | Skinny two-shadow cap at the root-delta scale, the corrected Route-B statement superseding the retired dual-localization contract.                             |
| <b>ex-hume</b>                   | proved-mod-audit         | Explicit 3-by-3 sharpness family for the exponent in <b>op-classical</b> ; inherited proof is not rigorous here.                                               |
| <b>lem-canonical-separator</b>   | proved-mod-audit         | Canonical separator package for a hidden top vertex and its harmonic deficit; inherited/mod-audit only.                                                        |
| <b>lem-dual-localization</b>     | obstruction (superseded) | Retired 2026-07-04: the transcribed contract is a distance tautology; superseded by <b>conj-skinny-shadow-cap</b> .                                            |
| <b>lem-exposed-circuit</b>       | proved-mod-audit         | Concentration and circuit estimate for exposed vertices; inherited/mod-audit only.                                                                             |

|                               |                                       |                                                                                                                                                                            |
|-------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lem-collateral-import         | proved (af:<br>validated)             | Collateral-import bound (CI) for the pivot-removing move, sharp on the wave-10 witness; af-validated in-repo 2026-07-04 (32-node tree, taint clean).                       |
| lem-cross-pivot-cancellation  | proved (af:<br>validated)             | Cross-pivot B-L cancellation identity behind the import anatomy; af-validated in-repo 2026-07-04 (23-node tree, taint clean); near-definitional.                           |
| lem-import-reduction          | proved (af:<br>validated)             | Universal reduction of the collateral import to the cross-pivot masses; af-validated in-repo (G11 reduction (4)); section shard pending.                                   |
| lem-leakage                   | proved-mod-<br>audit                  | Affine-face leakage estimate for almost-idempotent stochastic rows; inherited/mod-audit only.                                                                              |
| lem-kitaev-almost-idemp-audit | proved-mod-<br>audit                  | Corrected hostile-audited diagrammatic core of <code>th_almost_idemp</code> : dimension-free $10\eta$ associativity and the explicit extended interface; not af-validated. |
| lem-kitaev-diagonal-repair    | proved-mod-<br>audit                  | Finite phase-balanced repair of Kitaev's false direct-sum diagonal formula, with exact projective norm one and no block-count loss.                                        |
| cor-kitaev-diagonal-cpization | proved-mod-<br>audit                  | Entrywise complete-positivity corollary from the repaired exact central diagonal; exact multiplicativity is not assumed.                                                   |
| lem-prh                       | proved-mod-<br>audit                  | Positive-retract hardening with bound $2\sqrt{2\varepsilon}$ ; hostile-verified paper proof, not af-validated.                                                             |
| lem-prh-sharpness             | proved-mod-<br>audit                  | Two-scale PRH family forcing distance at least $\sqrt{\varepsilon}/2$ from every stochastic idempotent.                                                                    |
| lem-thmainext-conditional     | proved-mod-<br>audit<br>(conditional) | Conditional on <code>conj-hcb</code> and <code>conj-extcb</code> , the full amplified assembly and universal-constant ledger close; both premises remain open.             |
| lem-wiggle-rigidity           | proved-mod-<br>audit                  | F-WR wiggle-rigidity estimate for common-pattern webs; side conditions remain inherited.                                                                                   |
| obs-deep-leakage              | heuristic                             | Heuristic obstruction showing why a purely local row-identity argument cannot price deep-side mass.                                                                        |
| obs-fwr-gap                   | heuristic                             | F-WR forbidden-gap obstruction to using web rigidity as the dimension-free shallow-class counter.                                                                          |
| obs-linear-law-finite-delta   | numerical                             | Exact finite-delta certificate exceeding the empirical linear-law constant; L3 evidence, not proof.                                                                        |
| obs-orphan-amplifier          | proved-mod-<br>audit                  | Exact two-orphan family killing every class/signed-only orphan budget; repaired ratio tends to four; wave paper-proof, orchestrator-recomputed.                            |
| obs-sigma-halo-nonrobust      | numerical                             | Exact certificate that the literal zero-halo sigma cap is false; L3 evidence and obstruction.                                                                              |
| op-classical                  | open                                  | North-star classical projection stability problem; still open in this repo.                                                                                                |
| op-exposed-hull               | open                                  | Global exposed-hull lemma that would imply <code>op-classical</code> through the inherited factorization route; open.                                                      |
| prop-approx-simplex           | proved-mod-<br>audit                  | Approximate-simplex reduction to a stochastic idempotent; inherited/mod-audit only.                                                                                        |
| thm-classical-factorization   | proved-mod-<br>audit                  | Conditional classical factorization theorem through cluster geometry and <code>op-exposed-hull</code> ; not rigorous here.                                                 |
| thm-cluster                   | proved-mod-<br>audit                  | Cluster geometry reduction for signed affine retractions; inherited/mod-audit only.                                                                                        |

|                                   |                  |                                                                                                   |
|-----------------------------------|------------------|---------------------------------------------------------------------------------------------------|
| <code>thm-corner-constants</code> | proved-mod-audit | Corner-scale constants, with numerical components explicitly not promoted.                        |
| <code>thm-rank-one</code>         | proved-mod-audit | Rank-one signed affine retraction stability, including the Hume family; inherited/mod-audit only. |
| <code>thm-simplex</code>          | proved-mod-audit | Simplex and one-dimensional row-polytope stability reduction; inherited/mod-audit only.           |
| <code>thm-well-exposed</code>     | proved-mod-audit | Well-exposed-vertex route into the simplex theorem; inherited/mod-audit only.                     |

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